Project Title: 📦 Order & Shipping Insights Dashboard

Duration: 2–3 Hours

🎯 Objective

Using the provided order dataset, your task is to analyse return trends, product performance, and shipping method efficiency using NumPy, Pandas, Matplotlib, and Seaborn.

Dataset

order_shipping_data.csv (generate from AI tool)

Dataset Fields:

- Order_ID
- Customer_ID
- Product
- Category
- Order_Quantity
- Price
- Return_Status
- Shipping_Method
- Order_Date
- Total_Price

📌 Tasks Checklist

🔹 Data Preparation

1. Load the dataset and check for missing values.
2. Convert Order_Date to datetime format.
3. Create a new column for Week_Number and Month using Pandas datetime features.

🔹 Data Analysis

4. Find the top 3 most sold products.
5. Calculate the return rate (%) for each product.
6. Determine which shipping method has the highest return rate.
7. Group total revenue by:

- Product
- Shipping Method
- Return Status

🔹 Data Visualization

8. Plot a bar chart for total revenue by product.
9. Plot a pie chart showing distribution of shipping methods.
10. Create a line chart of total orders over time.
11. Use Seaborn to:

- Create a countplot of returns across categories.
- Create a violin plot of order quantity by product.

📝 Submission

Python script or Jupyter notebook with:

- Clean, commented code
- All tasks completed
- Clearly labeled graphs